

Project Report on

Training and Placement Cell System for Thakur College

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Introduction

Training and Placement Cell of TCSC is a total management and informative system, which provides the up-to-date information of all the students in thakur college. Training and Placement Cell System

helps overcome the difficulty in keeping records of hundreds and thousands of students along with searching for a student's eligibility based on recruitment criteria. It helps in co-ordination between Placement Cell officer and students. Maintaining student records, updating curriculum details and retrieving eligibility students list based on the company's criteria.

Supported by databases, Training and Placement Cell has a major role in TCSC and other colleges in which most of the work till now is carried out manually. The main aim of this project is to automate the Training and Placement procedure in our college. This project reduces manual work and maximizes the optimization, abstraction and security. This is a web application which will help students as well as the administration authority to carry out each and every activity in campus hiring.

This system is an application that can be accessed throughout TCSC and outside as well with proper login credentials. This system can be used as an efficient, quick way for the Training and Placement cell of the college to manage student information regarding placement recruitment.

In today's world everyone is travelling for jobs after completion of their graduation as it has become the need of each and every student, but for that they need to travel worldwide in searching of jobs.

The system is an online web application that can be accessed throughout the college with proper login provided. Student logging in should be able to upload their information like education details, Id proof, marksheets etc. to check for eligibility and other opportunities. College representatives logging in may also access or search any information put up by the students.

The cell have to collect the information and manage them manually according to various streams. If any modification is required that is also to be done manually.

In the system, the student's data is maintained in excel sheets. According to the company's requirement, the data is short listed manually by the cell as handling and maintaining the student's data manually is tedious to the cell and sometimes it might be inaccurate.

The main purpose of this project is to add new features to existing system. The proposed one is an online system which can be accessed throughout the college and outside by providing valid login credentials. This system can be used as an application for the Training and Placement Cell of the college to manage student's information regarding placements. The student's record includes personal details, educational qualifications, professional skills and academics, etc., This system acts as central repository for student information.

The use of the Internet and the World Wide Web has revolutionized the provision of information and the facility for the user to take action on the information obtained. Using the internet the cell can help students and companies to manage the placement process with the active involvement of the placement co-ordinator. This led to a unique web-based training and placement management system.

This system provides information on placement providers and the placements they offer so that students may view and assess their opportunities.

The users can access easily to this and the data can be retrieved easily in no time.

All transactions in the system are done manually in maintaining the records. The management and all the departments that have been carrying out this job manually makes the job more complicated and tedious most of the times. This laborious job can be made simple by computerizing the system.

Adding to this, college representatives can also post notices about internship programmes for students along with the registration forms which student can fill to avail the opportunity. Internships will help students get a solid industrial experience and boost their resume, helping them get better placements in the future.

Problem Statement

- Students choose a specific college where the placement will be held, there is a need to maintain all these papers, which leads to the requirement of large amount of space. It is manually done, chances of missing, difficult to handle the details of student.
- The existing system is doing all the processes manually. The administrator should refer to all the records kept for years ago to simply know details. This so tedious and time consuming. This process is so difficult when the number of users increases.
- In manual training and placement all the work that is done at TCSC is by human intervention due to which there are maximum chances of errors. The interface of student and the cell is maximum which makes the system time consuming.
- The students are not being made aware of the Training and Placement activity hence there might have been losing opportunities.
- The records are stored in modified access sheets hence sorting is a problem. The files are not stored hierarchical format hence searching is a big problem due to this the updation is very difficult and ambiguous leading to the duplication of records is usual hence data redundancy.

- There was no record kept of the past students. There is less communication between past or present student with the Training and Placement department.

Objectives

- **Efficient Resource Utilization:** Optimize the use of resources by automating processes, allowing the Training and Placement Department to focus more on strategic planning and student support.
- **Improved Transparency:** Enhance transparency in the placement process by providing clear visibility into eligibility criteria, selection procedures, and placement opportunities for both students and administrators.
- **Enhanced Accessibility:** Ensure easy access to placement-related information for students, providing a centralized platform to explore and apply for opportunities without physical constraints.
- **Feedback Mechanism:** Implement a feedback system where students can provide insights on their placement experiences, aiding in continuous improvement of the overall placement process.
- **Skill Development Tracking:** Introduce features to track and showcase the skill development journey of students, helping them better understand and demonstrate their capabilities to potential employers.
- **Industry Interaction:** Facilitate increased interaction between students and industry representatives through the platform, creating networking opportunities and industry-specific insights.
- **Adaptive Learning Modules:** Integrate adaptive learning modules to help students enhance their skills in areas identified as crucial for successful placements, promoting continuous learning and development.

Scope of Project

Our project has a big scope to do. Students can access previous information about placement.

This project covers wide scope of information of all the students which can be stored in the format of CV's that are further categorized according to various streams. Various companies can access the information of students and they can maintain and update their information. Notifications are sent to students about the companies.

It may help in collecting perfect management details, also a person accessing the system will understand about the working mechanism of the cell.

Students or college representatives logging in can get information about :

- Company Page
- Company recruitment schedule Page
- View Recruitment Schedule
- View Selected List of Students
- Coordinator page, ongoing page, process page.
- Upload technical papers.
- Analysis of overall placement activities

Limitations

- i. **Dependency on User Input:** The success of the system relies heavily on the timely and accurate input of data by students, faculty, and recruiters, which may be subject to errors or delays.
- ii. **Internet Dependency:** The system's functionality may be limited by the availability and reliability of internet connectivity, impacting users who may face network issues.
- iii. **Limited Access for Non-Tech Savvy Users:** Users with limited technical proficiency may find it challenging to navigate the system, potentially affecting their engagement and the overall effectiveness of the platform.
- iv. **Data Privacy Concerns:** Despite security measures, there is always a risk of data breaches or unauthorized access, raising concerns about the privacy and confidentiality of student and company information.
- v. **Resource Constraints:** Limited financial and technical resources may hinder the development, maintenance, and scalability of the system, impacting its long-term viability.
- vi. **Evolution of Placement Trends:** The system may struggle to adapt quickly to changes in job market trends or placement practices, potentially making some features less relevant over time.
- vii. **Resistance to Change:** Resistance from staff, students, or recruiters to adopt new technology and processes may impede the

- successful implementation and utilization of the placement cell system.
- viii. **Limited Feedback Mechanism:** The system might lack a robust feedback mechanism, making it challenging to gather insights from users for continuous improvement.
 - ix. **Mismatch with Industry Standards:** Changes in industry best practices for recruitment and training may outpace the system's ability to adapt, resulting in potential misalignment with current practices.
 - x. **Inadequate Training:** If the user training and support are insufficient, it may lead to underutilization of the system's capabilities and diminished overall effectiveness.
 - xi. **Limited Scope:** The system may focus primarily on placement activities, potentially neglecting broader career development aspects that could enhance the overall student experience.

Software Requirement Specification (SRS)

The Software Requirements Specification(SRS) provides a description of all the functions and constraints of the placement management system, developed for the TCSC placement cell. It will also manage the data of the company which would comprise of the profile of the company, eligibility criteria and the facilities or the package it provides etc. This system is specially designed to cater to the needs of the placement cell of TCSC , offering a centralized platform for both students and companies involved in the placement process.

The primary focus of the training and placement cell is to maintain a database that encompasses all student records, ranging from academic details to personal information.

The System would provide the facility of viewing both the personal and academic information of the student and company, it would also search for eligible students and company and deal with the insertion and deletion of records.

The intended audience of this document includes faculty members in the Department of IT, the developers and the students looking for On-Campus placements.

Information displayed, and other statistical information will attract new admission and a clear picture with records will be maintained with this portal.

Additionally, the document is designed to be accessible and informative for company HR personnel, offering a clear understanding of the college's placement processes. The placement cell, aims to not only assist in the placement process but also serve as a valuable tool for attracting new admissions. The portal ensures a transparent and organized presentation of records, contributing to a more efficient and effective management of the college's placement activities.

Performance Requirements

- The users must get the response within seconds. i.e. the response time of a particular function should be minimum.

Security Requirements

- The System software must be equipped with adequate security features to prevent data breach.
- There must be enough security to prevent unauthorized users logging in the system and maintain the privacy.

Functional Requirements

1. Feedback Mechanism: System for collecting feedback from students and employers about the placement process, including interview experiences and job satisfaction.

2. Interview Scheduling: Feature for scheduling interviews between students and employers, with options for both parties to view and confirm available time slots.

3. Job Posting: Ability for employers to post job vacancies, including job descriptions, requirements, and application deadlines.

4. Communication Tools: Integrated communication tools for sending notifications, updates, and reminders to students, faculty, employers, and other stakeholders.

5. Student Registration: Capability for students to register with the placement cell, providing necessary personal, academic, and professional details.

6. Generate Placement Reports: The system should provide functionality to generate detailed placement reports based on the data collected and managed by the training and placement cell.

Non-Functional Requirements

1. Performance

The system should be able to handle a large number of simultaneous users, ensuring quick response times during peak periods, such as registration or job application deadlines.

2. Compatibility

The system should be compatible with various devices, browsers, and operating systems to accommodate the diverse preferences and technologies used by students and employers.

3. Usability

The user interface should be user-friendly, ensuring that both students and employers can navigate the system easily without extensive training.

Modules Used

TCSC training and placement system provides the modules like :

- Student
- Admin
- Login
- Feedback
- Placement
- Training
- Company
- Data analysis
- Student can view company data
- Admin dashboard has overall functional rights
- Appropriate data processing and handling

Student Module

This module consists of a login option for registered students and a registration window for unregistered students. Students will use their choice of username and a proper specified length password. The functionalities provided in this module are:

- Allows Students to update their details anytime.
- The students would be able to view the company requirement.
- Students will be provided with a link to apply for the company if eligible.
- Previous Years placement paper and material can also be viewed.

- Students can see the complete profile of the company, such as recruitment procedure history, CTC offer and the working environment.
- Students can view for internship notices and fill registration forms accordingly.
- Students can fill feedback forms provided by the cell online.

Data analysis module

1. Introduction to Data-Driven Placement Strategies:

- Emphasizing the role of data analysis in optimizing placement processes.
- Connecting data-driven insights to improved decision-making.

2. Data Collection for Student Profiles:

- Identifying key parameters for student profiles (academic, extracurricular, etc.).
- Establishing systematic data collection methods.

3. Placement Process Optimization:

- Analysing the efficiency of the placement process.
- Identifying bottlenecks and areas for process optimization.

4. Feedback Analysis Loop:

- Establishing a feedback loop for continuous improvement.
- Analysing feedback data from students, recruiters, and other stakeholders.

5. Skill Gap Analysis and Training Recommendations:

- Conducting skill gap analyses to tailor training recommendations.
- Aligning training programs with identified skill gaps.

This module is designed to empower the Training and Placement Cell with data-driven insights to improve the overall effectiveness of student placement strategies.

Admin Module

TCSC has the authority to manage various functionalities of the system. This module will be handled by an Admin (say Training and placement Officer) who has the authority to:

- Add company details
- Provide authentication for registered students
- Add news feed
- Update company data. Overall records of the students will be presented over the portal like the data of all placed and unplaced students which will reduce the bottleneck of confusion among students.
- Contact companies and inform about placement drives.
- Contact companies and inform about placement drives.
- Student Certification Management
- Interview Scheduling System
- Company Review and Rating System
- Surveys and Polls Module
- Automated Email Notifications

Company Relations Module:

Database of Companies:

Functionality: Maintains a comprehensive database of potential employers. This includes information about the industries they belong to, their hiring preferences, and the types of roles they typically offer.

Industry Outreach:

Functionality: Actively reaches out to various industries to create partnerships and collaborations. This involves promoting the educational institution as a valuable source of skilled talent.

Communication and Networking:

Functionality: Establishes communication channels with representatives from companies. Engages in networking activities, such as industry events, seminars, and conferences, to foster strong relationships.

Company Visits and Presentations:

Functionality: Organizes visits by company representatives to the campus. Facilitates presentations and interactions between students and companies, allowing students to learn about potential employers.

Job Profile Matching:

Functionality: Analyzes the job profiles and skill requirements of companies. Matches these with the skills and preferences of the student population to ensure a better fit during placement activities.

Internship Opportunities:

Functionality: Collaborates with companies to secure internship opportunities for students. Internships serve as a valuable bridge between academic learning and practical industry experience.

Feedback Collection from Employers:

Functionality: Gathers feedback from companies regarding the quality of students, the effectiveness of placement processes, and any areas for improvement. This feedback informs future strategies.

Negotiation and Agreement Management:

Functionality: Manages negotiations with companies regarding placement terms, internship structures, and other collaborative initiatives. Maintains records of agreements and ensures compliance.

Login Module

The login module is crucial for managing user access. You'd typically need:

- 1. User Registration:** Allow students and administrators to create accounts.
- 2. User Authentication:** Verify login credentials for access.
- 3. Role-based Access Control (RBAC):**
Assign different roles (like student, admin) with corresponding permissions.
- 4. Password Recovery:** Provide a secure way for users to reset forgotten passwords.
- 5. Session Management:** Ensure secure and efficient handling of user sessions.

Always prioritize security, and perhaps implement features like two-factor authentication for added protection.

Feedback Module

- 1. Feedback Forms:** Design and implement customizable forms for collecting feedback from students, recruiters, and faculty members.
- 2. Anonymous Feedback Option:** Allow users to provide feedback anonymously to encourage honest opinions.

3. Feedback Categories: Categorize feedback to gather insights on specific aspects such as campus infrastructure, placement process, or overall experience.

4. Rating System: Include a rating system to quantify feedback and easily identify areas that need improvement.

5. Comments Section: Provide space for users to provide detailed comments or suggestions along with their ratings.

Ensure that the feedback module complies with privacy regulations and follows best practices for data security and confidentiality. Regularly review and act upon the feedback received to enhance the overall efficiency and effectiveness of the training and placement processes.

Training Module

Training module for a Training and Placement Cell project involves several key components:

1. Interview Preparation:

- Mock interviews to improve communication and presentation skills.
- Techniques for handling common interview questions and scenarios.

2. Soft Skills Development:

- Training on interpersonal skills, teamwork, and leadership.
- Communication workshops to enhance verbal and written skills.

3. Technical Skills Enhancement:

- Identification of technical skills in demand.
- Workshops, seminars, and hands-on sessions for skill development.

4. Industry Interaction:

- Organizing guest lectures by industry experts.
- Networking events to connect students with potential employers.

5. Placement Assistance:

- Guidance on applying for job opportunities.
- Support in coordinating interviews and placement drives.

Placement Module

Placement module TCSC training and placement cell project, consider the following features:

1. Mock Interviews:

- Incorporate a feature for conducting mock interviews to prepare students for real-world job interviews. Include feedback and suggestions for improvement.

2. Alumni Network Integration:

- Connect with the college's alumni network to provide mentorship opportunities, career advice, and networking events for current students.

3. Internship Opportunities:

- Include a section for posting and applying for internships, allowing students to gain practical experience in their field of study.

4. Student Registration:

- Allow students to register with their details, academic achievements, and career interests.

5. Communication Platform:

- Implement a messaging system for communication between students, companies, and the placement cell.

6. Resume Upload:

- Allow students to upload their resumes, making it easy for companies to review their profile.

System Configuration

1. Software Requirements

- Operating System: Windows / Linux
- XAMPP/WAMPP
- Web Browser
- HTML
- CSS
- PHP
- JavaScript

2. Hardware Requirements

- Processor : Intel p4 or later
- RAM : 512 MB or More
- Hard Disk : 40 GB or more

Individual Work Load

Sumant Vishwakarma

Worked on SRS, Admin Module and company module along with database creation and connectivity.

SRS

Purpose:

The TCSC placement system, designed to manage student and company data for placement purposes. It serves as a centralized

platform for students and companies, providing a clear and organized view of records.

Key Components

1. Student Records
2. Company Data
3. User Registration
4. Job Posting

Overall it not only assist in the placement process but also serve as a tool for attracting new admissions.

Such module are as follow:

Admin:

Purpose: Handle overall system administration, user management, access control, and system configuration.

Objectives:

- Control and monitor system activities.
- System security and integrity.
- Manage user roles and permissions.

Problem

Access Control: Significant of balance between providing sufficient access for system administration and ensuring security

Company:

Purpose: It Manage information about companies participating in the placement process, including profiles and job requirements.

Objectives:

Facilitate communication and collaboration with recruiting companies.
It provide information about job opportunities.

Problem:

Data Accuracy: Verifying and updating company profiles and job requirements regularly.

Atharva Dubal

Worked on Student and Data Analysis Module along with user interface and homepage of the TPC Website.

Student Module :

Purpose: Manage and organize information about students, including profiles, academic records, and resumes.

Objectives:

- Track and manage potential candidates for placements.
- Provide a centralized repository of student information for easy access.

Problems:

Data Accuracy: Ensuring that student information is accurate and up-to-date might be a challenge.

Privacy Concerns: Implementing robust data privacy measures to protect sensitive student information.

Data Analysis Module :

Purpose

- Focus on analytics, analysing data related to placements, student performance, and relevant metrics.

Objectives

- Make data-driven decisions.
- Identify trends to improve overall processes.

Problems

- Data Quality: Ensuring the quality and reliability of data for accurate analysis.

- Interpretation: Accurately interpreting data trends and making informed decisions based on analysis.

Sumit Soni

Worked on Login and Feedback Module along with Login Page and About Us Page of the TPC Website.

Login Module :

Purpose

- Handle authentication and authorization for secure access to the system.

Objectives

- Ensure only authorized users can log in.
- Grant access based on roles and permissions.

Problems

- Security Risks: Mitigating the risks associated with unauthorized access and potential security breaches.
- Password Management: Ensuring strong password policies and providing secure options for password recovery.

Feedback Module :

Purpose

- Collect feedback from stakeholders, such as students and recruiters, to evaluate and improve training and placement processes.

Objectives

- Gather insights into training program effectiveness.
- Enhance the overall quality of placement experiences.

Problems

- Feedback Collection: Encouraging active participation and obtaining honest feedback from stakeholders.
- Analysis and Action: Efficiently analyzing feedback and implementing actionable improvements.

Ritesh Gupta

Worked on Training and Placement Module along with Report Generation and Aptitude Test.

Training Module:

Purpose

- Organize and manage training programs, workshops, and skill development sessions.

Objectives

- Enhance the skills of students for better employability.
- Track attendance and assess the effectiveness of training sessions.

Problems

- Attendance Tracking: Ensuring accurate attendance tracking for training sessions.
- Relevance of Training: Adapting training programs to meet the evolving needs of the job market.

Placement Module :

Purpose

- Manage the placement process, including scheduling interviews, maintaining placement records, and coordinating with companies.

Objectives

- Ensure a smooth placement experience for students.
- Maintain accurate records of placements.

Problems

- **Coordination Challenges:** Managing the logistics of scheduling interviews and coordinating with numerous companies.
- **Communication Gaps:** Ensuring effective communication between the placement cell, students, and companies.

Process Development model used

Agile Model

The Agile development model is an iterative and incremental approach to software development that prioritizes flexibility, collaboration, and customer feedback. It emphasizes delivering working software in small, frequent releases, adapting to changing requirements, and continuously improving the development process.

The Agile Model is used in our project as it has the following advantages:

- ❖ **Flexibility in Requirements:** Training and placement activities often involve dynamic requirements, as student preferences, employer needs, and industry trends can change rapidly.
- ❖ **Continuous Feedback:** In the context of a training and placement cell, this means that students, faculty, employers, and administrators can provide feedback throughout the development process, ensuring that the system meets their needs effectively.
- ❖ **Adapting to Changes:** The job market and student needs change a lot. Agile helps us to adjust our plans quickly when things change, so we can keep up with what students and employers need.

- ❖ **Doing What Matters Most First:** We focus on doing the most important things first. For example, we make sure students have tools to build their resumes and get ready for interviews before adding extra features.

Agile helps us work together efficiently, make sure we're building the right things, and quickly adapt to changes so we can help students succeed in finding jobs.

Gantt Chart

A Gantt chart is a chart that displays a timeline of events and the associated tasks, milestones, or activities. Gantt charts are used in project management to identify task and project start dates, durations, end dates, and more.

Gantt charts can be used on any size of project, whether big or small, or simple or complex.

Steps of making a Gantt chart:

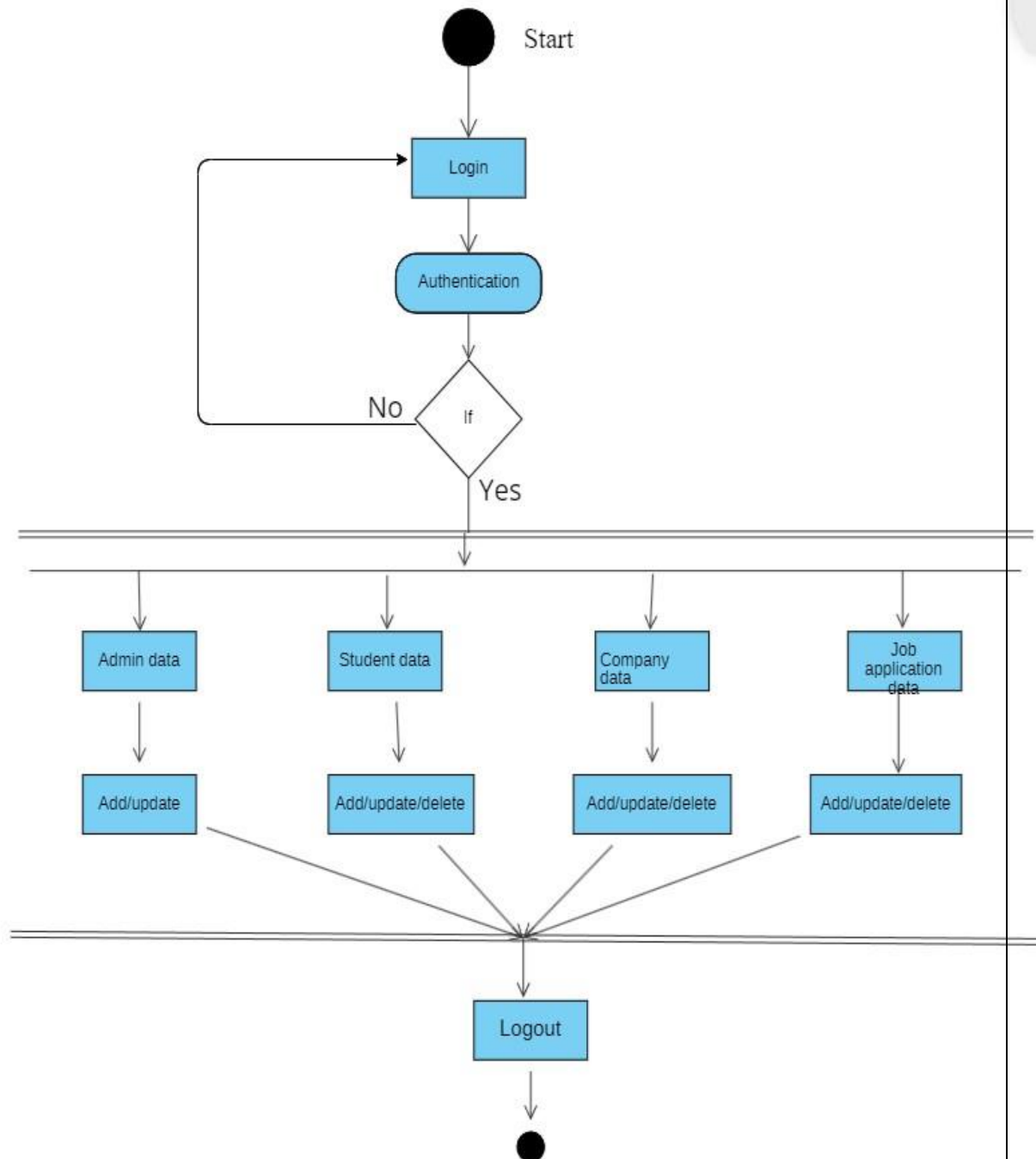
1. Identify Tasks and Activities in the Project
2. Task Durations
3. Choose a Gantt Chart Tool
4. Enter Task Data
5. Format the Gantt Chart
6. Review and Adjust
7. Update as Needed



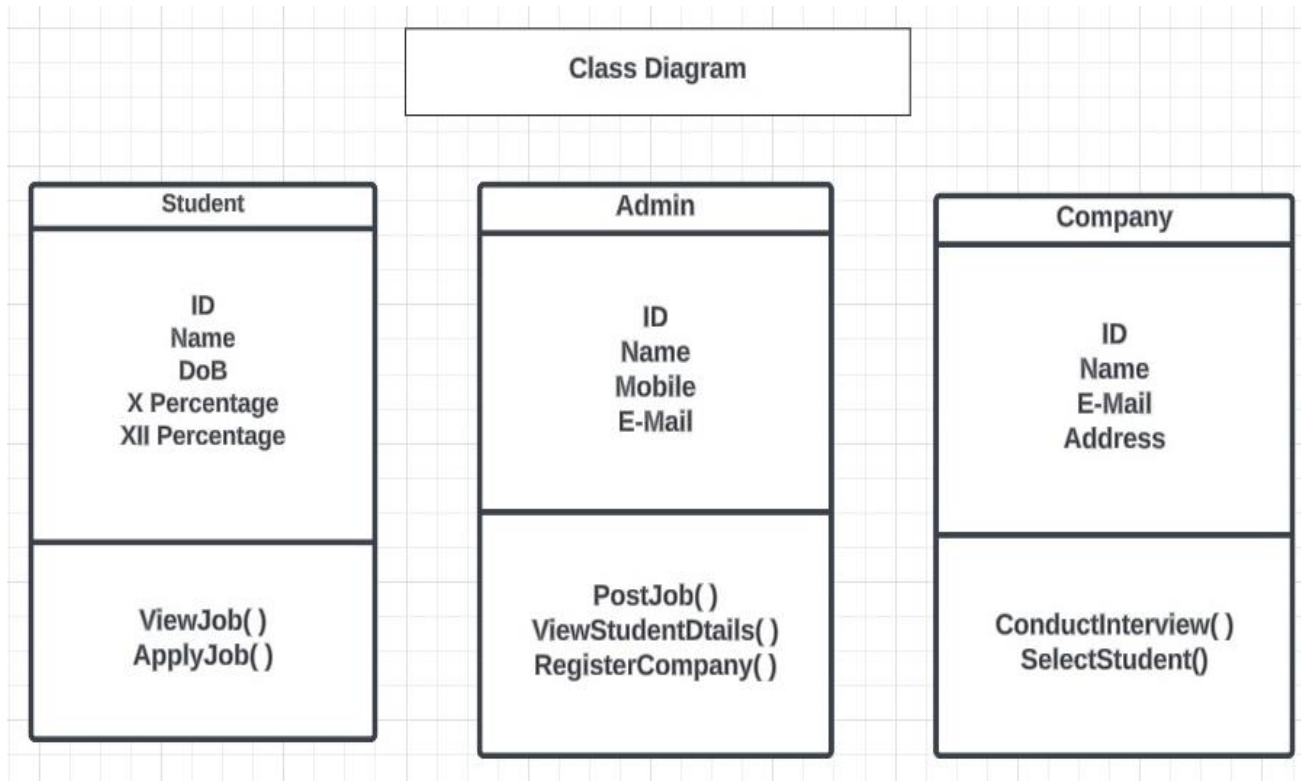
Project Schedule

SN	Activity	Date	Duration(Hrs)
1	Discuss & understanding	15-12-2023	1.5
2	Research	22-12-2023	1.5
3	Requirement & Analysis	24-12-2023	1
4	Module Distribution	28-12-2023	1
5	Documentaion	03-01-2024	1.5
6	Completion of PPT	04-01-2024	2
7	Meeting for Revaluation of Things	06-01-2024	1.5
8	Meeting for Analysing & Integration of Final Work	07-01-2024	1
9	Context Diagram	19-01-2024	2
10	DFD Diagram	19-01-2024	2
11	Use Case Diagram	02-02-2024	2
12	Class Diagram	02-02-2024	2
13	Er-Diagram	09-02-2024	2
14	Component Diagram	23-02-2024	2
15	Collaboration Diagram	23-02-2024	2
16	Activity Diagram	23-02-2024	1
17	Quality Management	25-02-2024	1
18	Risk Management	25-02-2024	1
19	Testing	29-02-2024	1

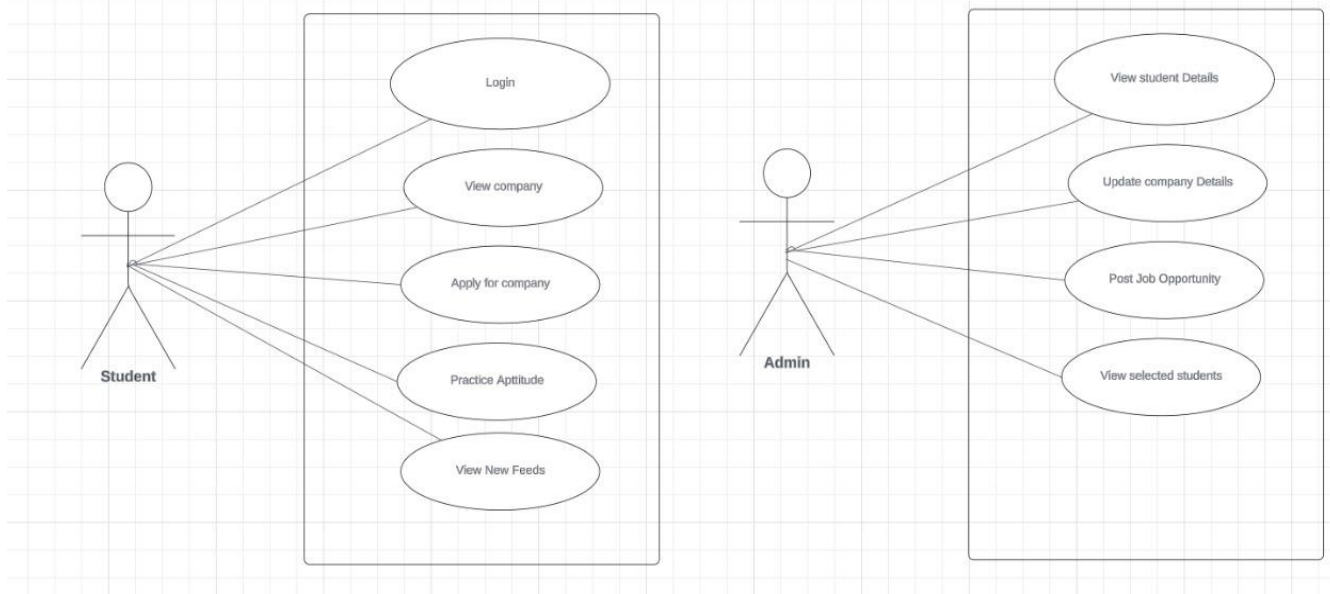
Study and Implementation of Activity Diagram



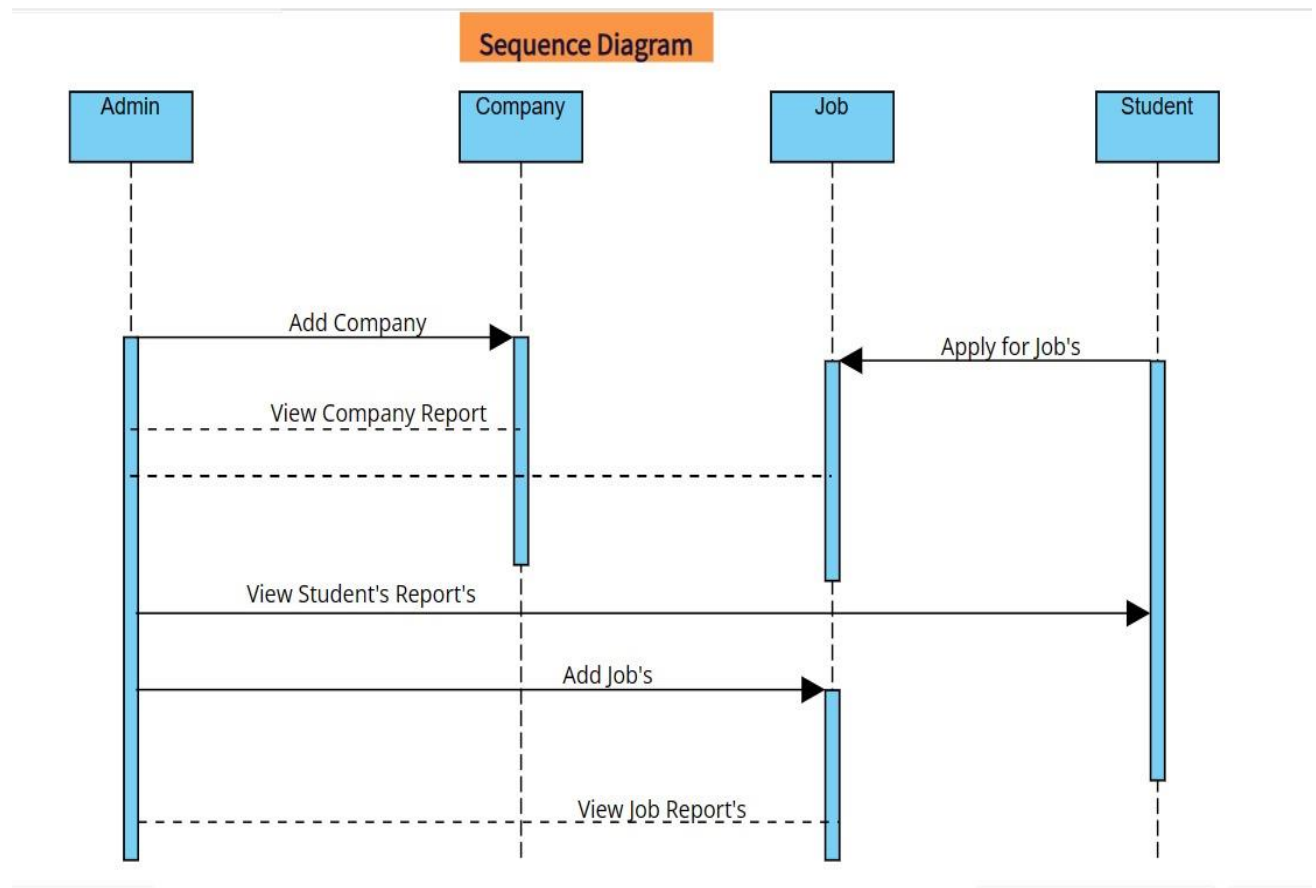
Study and Implementation of Class Diagram –



Study and Implementation of Use Case Diagram –



Study and Implementation of Sequence Diagram –



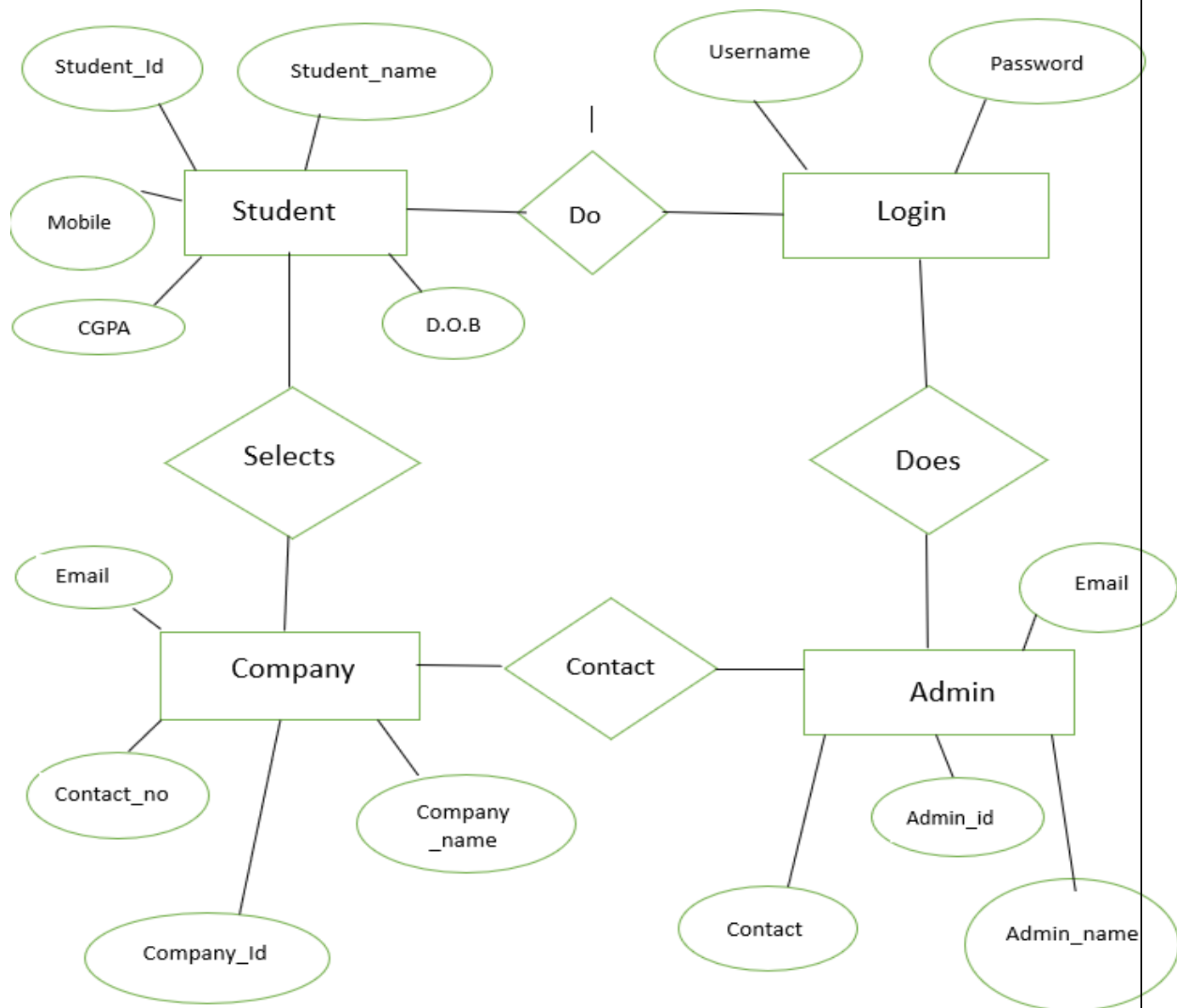
Study and Implementation of Entity – Relationship Diagram–

ER Diagram

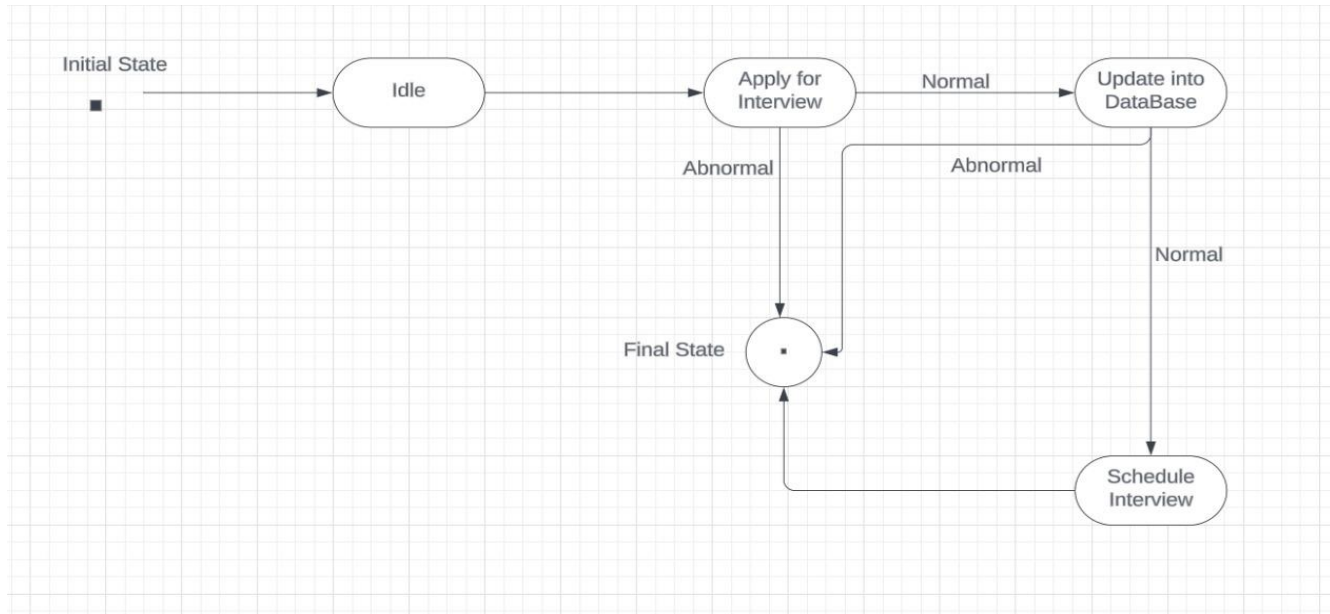
The Entity Relationship Diagram explains the relationship among the entities present in the database. ER models are used to model real-world objects like a person, a car, or a company and the relation between these real-world objects. In short, the ER Diagram is the structural format of the database.

Steps of making a Er-Diagram:

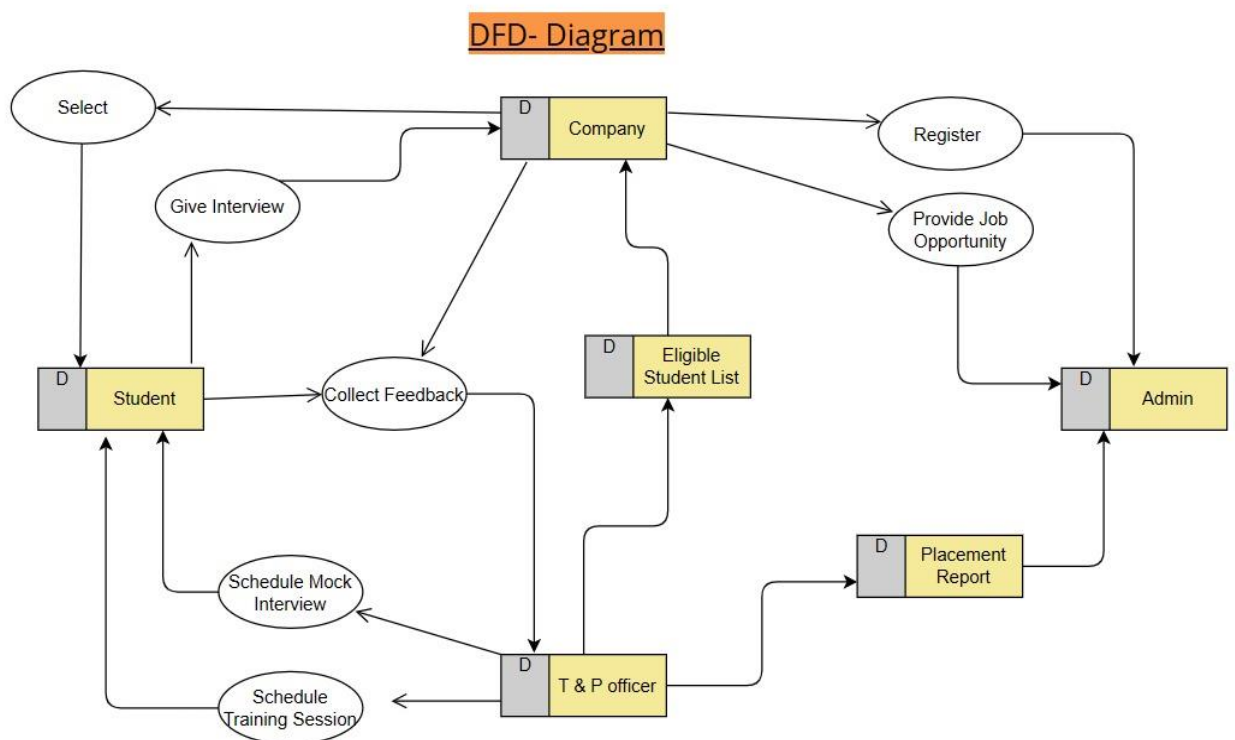
1. Identify Entities
2. Define Attributes
3. Establish Relationships
4. Specify Primary Keys
5. Draw Entities and Relationships
6. Label Relationships
7. Update as Needed



Study and Implementation of State Transition Diagram

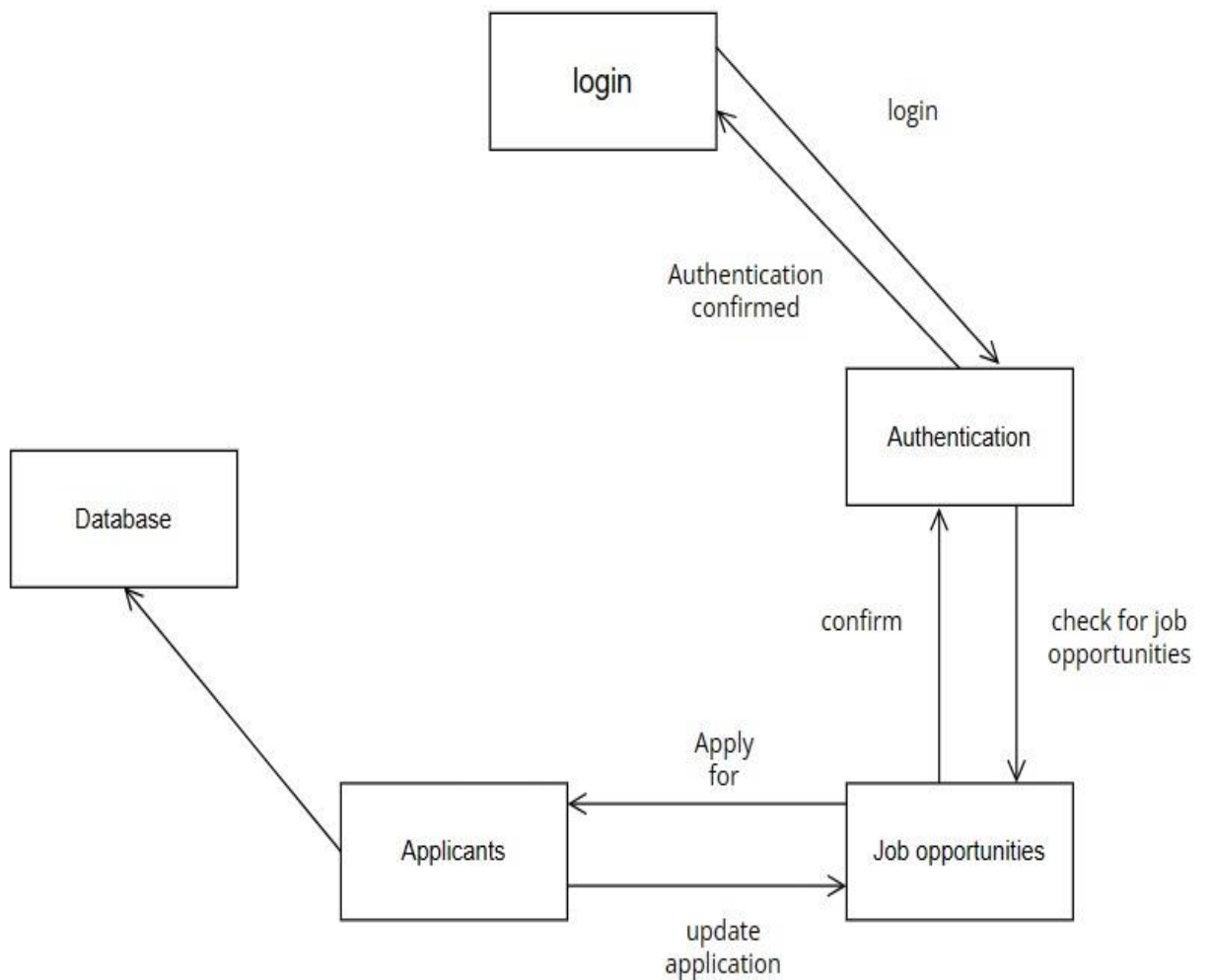


Study and Implementation of Data Flow Diagram –

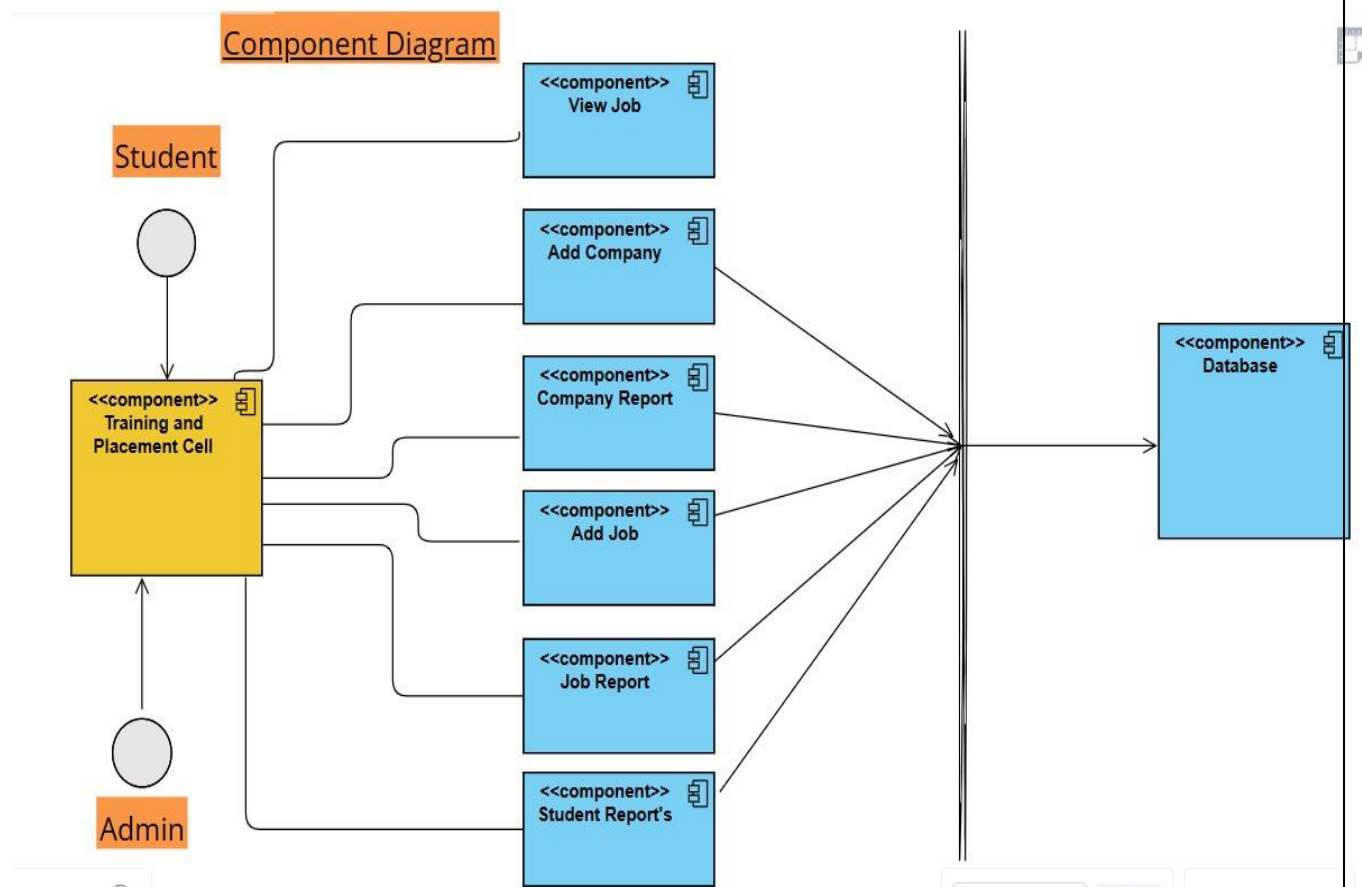


Study and Implementation of Collaboration Diagram –

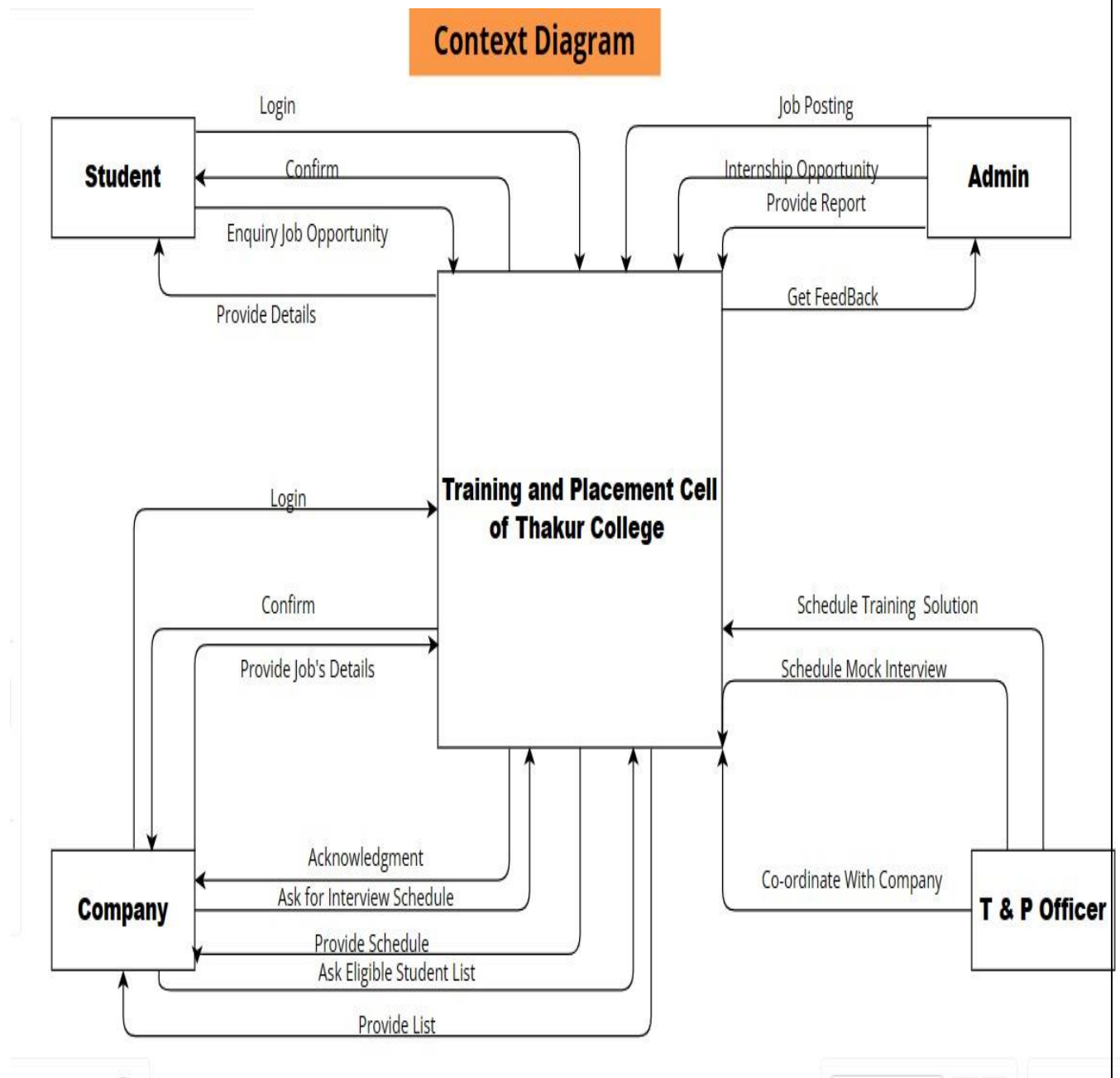
Collaboration Diagram



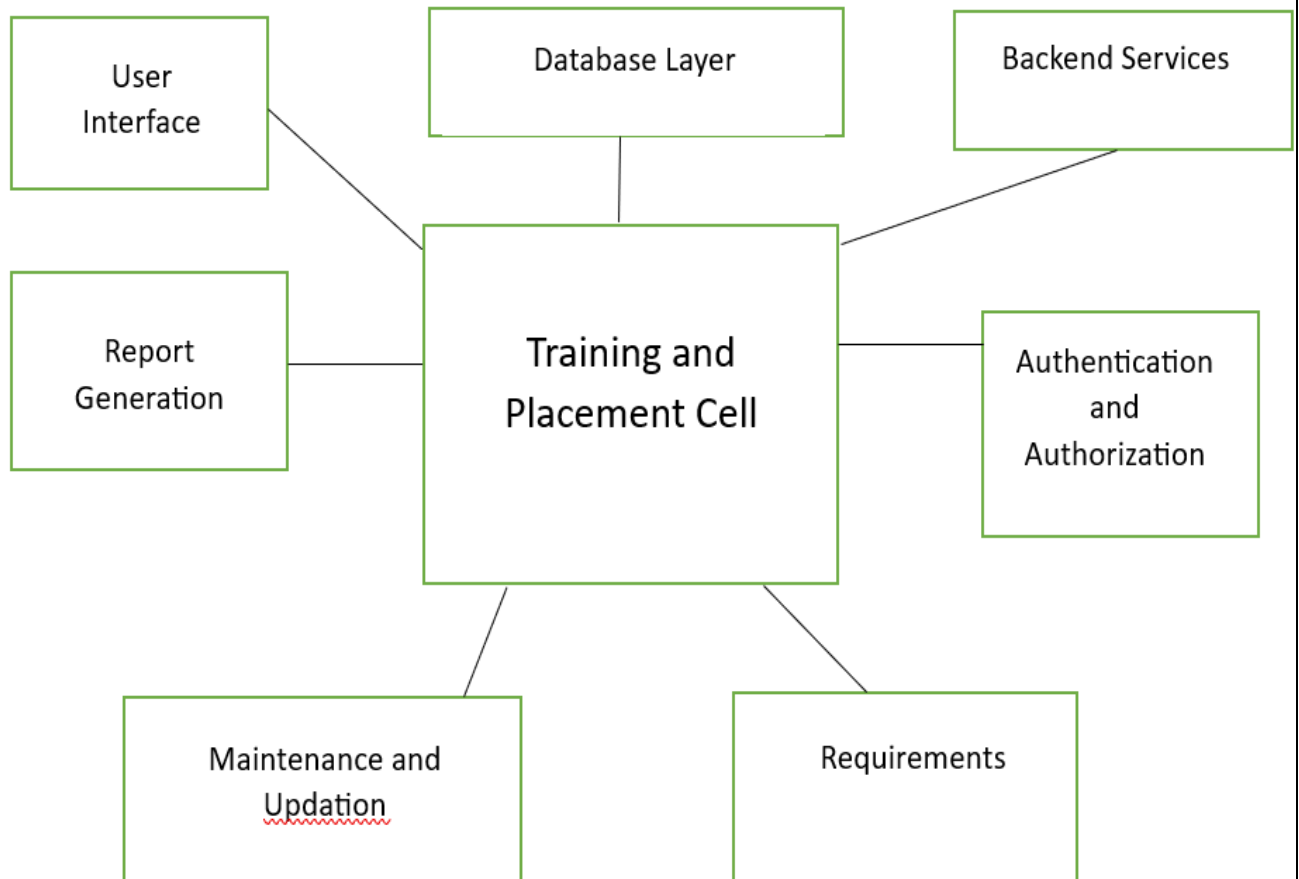
Study and Implementation of Component Diagram –



Study and Implementation of Context Diagram –



Architecture Diagram



Quality Management and Quality Plan

1. Define set of Quality Standards:

Establish clear quality standards and criteria for the placement process, including the quality of career counseling, resume preparation, interview guidance, and placement assistance.

- **Accuracy:**

In the context of a training and placement cell, accuracy ensures that information provided to students, employers, and faculty members is correct and reliable. This includes accurate job descriptions, student profiles, placement statistics, and other relevant data.

- **Quality Assurance Checks:**

Conduct regular quality assurance checks and audits to monitor the effectiveness and adherence to quality standards within the training and placement cell. Address any deficiencies or areas of concern promptly.

2. Process of Assessment :

The process of assessment in training and placement involves evaluating various aspects of students' skills, competencies, and readiness for employment, as well as assessing the effectiveness of placement services provided by the training and placement cell.

Implement continuous improvement initiatives to enhance the effectiveness and efficiency of placement processes.

- **Identify Assessment Criteria:**

Determine the key criteria for assessing students' employability, including technical skills, soft skills, communication abilities, problem-solving skills, and professional demeanor. Additionally, identify criteria for evaluating the effectiveness of placement services, such as placement rates, employer satisfaction, and student feedback.

- **Design Assessment Tools:**

Develop assessment tools and instruments tailored to the identified criteria. This includes skills assessments, aptitude

tests, mock interviews, behavioural assessments, and surveys for collecting feedback from students and employers.

- **Organize Mock Interviews:**

Conduct mock interviews to simulate real-world interview scenarios and assess students' performance in answering interview questions, articulating their qualifications, and demonstrating confidence and professionalism. Provide constructive feedback and coaching to help students improve their interview skills.

3. Quality Goals

Quality goals are specific, measurable objectives that an organization sets to achieve and maintain a high level of quality in its products, services, processes, or operations.

- **Increase Student Placement Rate:**

Increase the percentage of students placed in relevant jobs or internships within six months of graduation by X% compared to the previous year.

- **Enhance Student Satisfaction:**

Increase student satisfaction with placement services, as measured by satisfaction surveys, focus groups, or feedback forms, to achieve a rating of at least X out of 5.

- **Reduce Placement Process Cycle Time:**

Decrease the average time taken to match students with job opportunities and facilitate placements by X days or X%.

4. Risk Management:

Identify potential risks and uncertainties that may impact the quality of placement services and develop mitigation strategies to address them.

Regularly assess and monitor risks related to factors such as changing market conditions, technology disruptions, and regulatory compliance.

Risk management involves identifying potential challenges or uncertainties that could impact the successful placement of students into jobs or internships, as well as the effectiveness of training and placement activities.

- **Mismatch of Skills with Industry Demand:**

Risk: Students may lack the skills or qualifications required by employers, leading to difficulties in securing placements.

- Limited Employer Partnerships:

Risk: Dependency on a small number of employers for placement opportunities can limit students' options and increase competition.

- Limited Internship Opportunities:

A shortage of internship opportunities may hinder students' ability to gain practical experience and enhance their resumes.

5. Process Description

A process description outlines the steps, activities, and tasks involved in completing a specific process or workflow.

Process for Training and Placement Cell :

- Student Registration:

Students register with the training and placement cell to access placement services.

Collect student information, including personal details, educational qualifications, skills, and career interests.

- Job Posting and Matching:

Collect job postings from employers and organizations seeking to recruit students for internships, co-op programs, or full-time positions.

- Interview Preparation:

Conduct mock interviews to simulate real-world interview scenarios and help students practice answering common interview questions, articulating their qualifications, and presenting themselves professionally.

Risk management

- Identifying Risks:

- Identify potential risks that could affect the training and placement process, such as:
 - Insufficient industry demand for certain skills
 - Lack of student readiness for the job market

- Assessing Risks:

Assess the likelihood and potential impact of identified risks on the training and placement process.

- Mitigating Risks:

Develop strategies to mitigate or address identified risks, such as:

- Enhancing career counseling and skills development programs to better align with industry demand.
- Providing additional resources and support for students to improve their employability skills.

- Financial Risks:

Manage financial risks associated with training and placement activities, such as:

- Budget overruns or funding shortages impacting the availability of resources for placement services

Testing

Testing the project in the context of a training and placement cell involves evaluating the functionality, usability, and performance of the systems, processes, and services implemented within the project.

Unit Testing –

Test individual components or modules of the TPC in isolation to ensure they function correctly .Verify that each unit performs as expected and meets specified requirements.

Integration testing –

Integration Testing in a Training and Placement Cell project involves combining individual units or modules of the application and testing them as a group. This phase of testing is crucial for identifying issues related to the interaction between different parts of the system, such as data flow, dependency injection, and service integration. The goal is to ensure that combined units function together as intended.

System Testing:

Test the entire Training and placement cell as a complete system to validate its compliance with functional requirements and business objectives. Verify end-to-end functionality, including user workflows, data processing, reporting, and system performance.

User Acceptance Testing (UAT):

Involve end-users, such as admin, student , in testing the TPC to ensure it meets their needs and expectations. Conduct UAT in a production-like environment to assess usability, workflow efficiency, and user satisfaction.

Performance Testing:

Evaluate the performance and scalability of the Training and placement cell under various load conditions, such as peak usage times or increased data volumes. Identify and address any performance bottlenecks or scalability limitations that may impact system performance under heavy loads.

Approach of Testing :

Black – Box Testing –

Black-box testing is a software testing technique where the internal workings or structure of the system being tested are not

known to the tester. Instead, the tester interacts with the system's inputs and outputs, focusing on validating its functionality and behaviour based on specified requirements or expected outcomes.

BLACK-BOX TESTING

1)Error in Interface

- Check's if the interface has all the required modules

2)Link working

- use to test if we can redirect to different pages from the dashboard

3)Error with database

- use to validate, check and update the database information

WHITE-BOX TESTING

Applying white box testing in a Training and Placement Cell project involves a deep dive into the internal workings of the project's codebase. This type of project typically features several modules such as student registration, company onboarding, job postings, application submissions, interview scheduling, and result notifications.

1. Code Coverage Testing: Ensure that all parts of the code are executed during testing, including statements, branches, and paths.

2. Control Flow Testing: Verify that the control flow of the program behaves as expected, including loops, conditionals, and function calls
3. Error Handling Testing: Check how the system responds to errors and exceptions, ensuring graceful degradation and proper error messages.

Testable Units :

Login
News Feed
Practice Aptitude test
Feedback System
Placement Report Generation
Student Registration
Company Registration
Update Student profile
Application Submission

1. Login

Login Function:

Tests can verify if the login function correctly authenticates users based on valid credentials and rejects invalid ones.

Database Insertion:

Verifies that student data is correctly inserted into the database with proper handling of any constraints.

2.News Feed

Updation of news feed :

Admin can update news and post in on the feed when needed.

Database Updation :

The feed must be updated in the database and stored for future purposes.

3.Practice Aptitude Test

Aptitude test for practice :

Students can give mock tests for practice.

4.Feedback System

Checking if the feedback system for students and recruiters works properly.

5.Placement report generation

Report Generation:

Tests the generation of reports (e.g., placement statistics) to ensure they correctly aggregate and display data.

Data Export:

Verifies that data export features (e.g., exporting applicant lists to Excel) work correctly and match the displayed information.

6.Student Registration:

- Verify that a student can successfully register for the platform.
- Ensure that all required fields (such as name, email, contact number, etc.) are mandatory.

7.Company Registration :

Company Profile Creation:

Tests to ensure that company profiles are created correctly, with all required information being validated and stored.

8.Update Student Profile :

Student Profile Update:

Confirm that students can update their profiles with new information (such as skills, education details, resume upload, etc.).

Ensure that changes reflect correctly in the database.

9. Application Submission

Application Process:

Verifies that applications to job postings are correctly submitted, including validation of any prerequisites (e.g., student CV upload).

Application Status Update:

Tests the functionality that allows changing the status of applications (e.g., from "submitted" to "under review").

Unit Testing (Test Cases)

1.Login

Case1:

Username correct but password wrong:

- Username: Ritesh
- Password: abc

Expected output:

Username and password wrong

Actual output:

Username and password wrong

Case2:

Username wrong but password correct:

Username: abc

Password:12345

Expected output:

Username and password wrong

Actual output:

Username and password wrong

Case3:

Username and password both are wrong:

Username: abc

Password: xyz

Expected output:

Username and password wrong

Actual output:

Username and password wrong

Case 4:

Username and password both are correct:

Username: Ritesh

Password:12345

Expected output:

Login successful(open home page)

Actual output:

Login successful(open home page)

2.Feedback System

Case 1 :

Feedback not being submitted :
Expected output :
Feedback submitted successfully.
Actual output :
Feedback not saved.

Case 2 :

Feedback submitted successfully :
Expected output :
Feedback submitted successfully.
Actual output :
Feedback submitted successfully.

3.News Feed

Case 1 :

News updation in the feed :
Input – Admin trying to update the news in feed.
Expected output :
Updated news visible in feed.
Actual output :
Updated news visible in the feed.

Case 2 :

News Feed not updated in database :
Input – Admin updates news in feed, feed updated but not updated in database.

Expected output :

Updated feed visible in database.

Actual output :

Updated feed not visible in database.

Case 3 :

News updated in both Feed and Database :

Expected output :

Updated news visible in feed as well as database.

Actual output :

Updated news visible in feed as well as database.

4.Placement Report generation

Case 1 :

Export Placement Report :

Description – Admin should be able to export/download the report in formats such as .CSV, .PDF.

Expected output :

Admin able to export report in various file formats.

Actual output :

Admin able to export report in various file formats.

Case 2 :

Filter options in placement report :

Description – Admin should be able to filter options in the report like date etc.

Expected output :

Admin able to filter options in the report.

Actual output :

Admin not able to filter options in the report.

Case 3 :

Successful placement report generation with filters, export of report in file formats :

Expected output :

Placement report generated and filters can be applied, report can be exported in various file formats.

Actual output :

Placement report generated and filters can be applied, report can be exported in various file formats.

5.Student Registration :

Case 1 :

Student registration details should be stored in database :

Expected output :

Student details will be stored in the database.

Actual output :

Student details are stored in database.

Case 2 :

Student should provide the same password in both password and confirm password fields, otherwise new user will not be registered.

Expected output :

Student should be registered only if password and confirm passwords fields match.

Actual output :

Student is only registered when password and confirm password fields match.

6.Company Registration

Case 1 :

Company details not updated in database

Expected output :

Company details updated in database and they are visible.

Actual output :

Company details updated in database and visible.

Case 2 :

Company details updated but not visible to students.

Expected output :

Students can view company details added.

Actual output :

Company details added are not visible to students.

Case 3 :

Company details updated in database and visible to students

Expected output :

Company details stored in database and also visible to students.

Actual output :

Company details stored in database and also visible to students.

7.Update Student Profile

Description – Student should update or edit his/her profile at any time without errors or difficulties.

Case 1 :

Student Details should be updated in database :

Expected output :

Student updated details visible in database.

Actual output :

Student details updated visible in database.

8.Application Submission

Case 1 :

Application should be submitted after filling :

Expected output –

Application submitted successfully.

Actual output –

Application submitted successfully.

Case 2 :

The eligibility criteria should be checked properly.

Expected output –

The eligibility is checked properly, only eligible students can apply.

Actual output –

The eligibility is checked properly, only eligible students can apply.

Case 3 :

The application should be stored in the database.

Expected output –

The application is being stored in the database, and is visible to admin.

Actual output –

The application is being stored in the database, and is visible to admin.

9. Practice Aptitude Test

Description – Students will get aptitude tests for practice based on past papers.

Case 1 :

The practice test questions should be properly visible to the students.

Expected output –

Student can view the practice test questions.

Actual output –

Student can view the practice test questions.

Case 2 :

The test should be submitted and student should be able to check the scores.

Expected output –

Student can submit and check score.

Actual output –

Student can submit and check score.

Conclusion

The main aim of the Training and Placement cell system is to reduce the communication gap between placement officer and placement interested students. Students can update their information effortlessly and independently. Also, the users can access the system online easily at any time. This reduces the manual work and tedious tasks which before, were done manually. Since majority of users have their personal cellphones, Training and Placement Cell system is developed so everyone can access it through the web browser. This web-based application is very user friendly, secure and easy to access by all authorized members.